

Bhoomi Suraksha — AI-GIS Hazard Red-Zone & Relocation Platform

Organization	Ministry of Home Affairs — National Disaster Response Force (NDRF), DM Division
Category / Theme	Software — Disaster Management
Requirement	An AI-GIS decision-support platform (dynamic mapping + scoring + prioritization) — not framed as an autonomous agent

Solution

Bhoomi Suraksha is an AI-driven GIS decision-support platform that gives the Ministry of Home Affairs and State Disaster Management Authorities one continuously updated picture of risk, instead of a static map redrawn only after the next disaster.

It continuously fuses satellite and terrain data — slope, rainfall intensity, land use, historic disaster records — into multi-hazard Red Zones covering landslide, flood, coastal erosion, and cloudburst, refreshed as new data comes in rather than fixed at initial survey time. Candidate relocation sites are scored for real carrying capacity — terrain, land use, water availability, infrastructure access — so a habitation is never moved from one unsafe or unviable location to another.

Hazard severity, population exposure, and disaster history are then combined into a ranked list of immediate, short-term, and medium-term relocation priorities, surfaced through a map-based dashboard built for a disaster-management official, not a GIS specialist. This is deliberately not an autonomous agent — the brief asks for dynamic hazard modelling and prioritization, so the engineering effort goes into getting the scoring right rather than into a tool-use framework the problem doesn't call for.

At a Glance

Layer	Component
Hazard data	ISRO Bhuvan / NRSC Landslide Atlas, SRTM/Cartosat DEM, IMD rainfall data, historic disaster records.
Scoring engine	GIS multi-criteria model (AHP weighted overlay, optionally blended with an ML susceptibility model) — a proven approach in India for site-suitability problems.
Population layer	Census/SECC habitation-level data overlaid on hazard scores to compute actual exposure, not hazard alone.
Carrying capacity	Suitability scoring of candidate relocation sites on slope, land use, water access, and infrastructure distance.
Dashboard	Map-first web dashboard showing Red Zones, ranked priorities, and suggested viable sites — no GIS expertise needed to read it.

Impact

- **National scale:** roughly 12.6% of India's land area is landslide-prone and over 40 million hectares are flood-prone — this gives every State DM Authority one continuously updated risk picture.
- **Reactive to proactive:** moves relocation ahead of the disaster in exactly the remote Himalayan, Northeastern, and Western Ghats habitations NDRF already struggles to reach quickly once disaster strikes.
- **Defensible prioritization:** gives states an evidence-based basis to allocate scarce relocation budgets, instead of ad hoc sequencing.

- **Reusable nationally:** the same scoring architecture applies to any state or hazard type by swapping in local terrain, rainfall, and population layers.

Why It Stands Out

- **No GPU or live model serving required:** the core is GIS plus multi-criteria scoring, runnable on a laptop CPU — a structurally lower risk profile than an LLM-serving stack that needs a GPU to even boot.
- **Built on real, accessible public data:** ISRO Bhuvan, the NRSC Landslide Atlas, open DEM data, and Census/SECC records already exist, so there's no hidden data-availability gap to work around.
- **Instantly readable demo:** a live map where clicking a district shows Red Zones and a ranked relocation list is an immediate, visual result judges don't need a technical briefing to understand.
- **Lower, well-trodden build risk:** GIS + AHP/ML scoring is a proven, testable pattern buildable in stages — lower and more predictable risk than a live agent stack or an LLM answering on data that barely exists.